A COMPOSITION OF BIRATIONAL MORPHISMS IS BIRATIONAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-morphisms/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot. Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Editorial note. Read in full under the Lead's 12:4x rule (screening hit re-read, pb 1475B). The author writes verbatim 'There is an immediate reduction to the case where X and Y are irreducible which we omit.' - the reduction is to the standard irreducible case and the rest of the proof is carried out on the minimal primes; recorded here as the author's own wording.

Allowed standard prerequisites (editorial).

- Birationality described through the unique minimal primes and the induced extension of function fields. Lemma \ref{lemma-birational-integral} and the surrounding discussion, used at proof lines 13087-13102. Gives the criterion applied to both maps and to their composition.

Source reference. Original lemma 0BAC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14205–14205. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0BAB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14204–14204. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $f : X \to Y$ be a birational morphism of schemes having finitely many irreducible components over a base scheme S. Assume one of the following conditions is satisfied*

(1) *f is locally of finite type and Y reduced,*

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(2) *f is locally of finite presentation.*

Then there exist dense opens $U \subset X$ and $V \subset Y$ such that $f(U) \subset V$ and $f|_U : U \to V$ is an isomorphism. In particular if X and Y are irreducible, then X and Y are S-birational.

Proof. There is an immediate reduction to the case where X and Y are irreducible which we omit. Moreover, after shrinking further and we may assume X and Y are affine, say $X = \mathrm{Spec}(A)$ and $Y = \mathrm{Spec}(B)$. By assumption A, resp. B has a unique minimal prime $\mathfrak{p}$, resp. $\mathfrak{q}$, the prime $\mathfrak{p}$ lies over $\mathfrak{q}$, and $B_\mathfrak{q} = A_\mathfrak{p}$. By Lemma 0BAB we have $B_\mathfrak{q} = A_\mathfrak{q} = A_\mathfrak{p}$.

Suppose $B \to A$ is of finite type, say $A = B[x_1, \ldots, x_n]$. There exist a $b_i \in B$ and $g_i \in B \setminus \mathfrak{q}$ such that b_i/g_i maps to the image of x_i in $A_\mathfrak{q}$. Hence $b_i - g_i x_i$ maps to zero in $A_{g_i'}$ for some $g_i' \in B \setminus \mathfrak{q}$. Setting $g = \prod g_i g_i'$ we see that $B_g \to A_g$ is surjective. If moreover Y is reduced, then the map $B_g \to B_\mathfrak{q}$ is injective and hence $B_g \to A_g$ is injective as well. This proves case (1).

Proof of (2). By the argument given in the previous paragraph we may assume that $B \to A$ is surjective. As f is locally of finite presentation the kernel $J \subset B$ is a finitely generated ideal. Say $J = (b_1, \ldots, b_r)$. Since $B_\mathfrak{q} = A_\mathfrak{q}$ there exist $g_i \in B \setminus \mathfrak{q}$ such that $g_i b_i = 0$. Setting $g = \prod g_i$ we see that $B_g \to A_g$ is an isomorphism. □